

Session 2: Data Acquisition for NextGen and T-Route

Step-by-step commands for generating the river network and meteorological forcing data (subset, forcings, and realization) using the `ngiab-prep` tool. Two workflows are documented: a native Linux/conda environment and a Docker container.

Workflow A: Native Linux (conda environment)

These steps assume you are inside the activated conda environment.

Step 1: Install uv and review ngiab-prep options

Install the `uv` package manager, then display the `ngiab-prep` help to review the available options and flags:

```
pip install uv
uvx ngiab-prep -h
```

The help output lists all subsetting, forcing-generation, and realization options, including the model-selection flags (`--lstm`, `--summa`, `--snow17`, `--sacsm`, etc.), the data source (`--source {aorc,nwm}`), and the subset type (`--subset_type {nexus,catchment}`).

Step 2: Set the default output directory

Point `ngiab-prep` at the directory where future outputs should be stored:

```
uvx ngiab-prep --output_root /home/msmg8/Workshop/data/
```

Step 3: Run subset, forcings, and realization

Run the combined operation for the Coralville gage:

```
uvx ngiab-prep -i gage-05454300 -sfr \
--start_date 2017-01-01 --end_date 2018-12-31 \
-o Coralville --source aorc
```

Flag reference for this command:

- `-i gage-05454300` : input feature, given as a USGS gage ID.
- `-sfr` combined flags: `-s` subset the hydrofabric, `-f` generate forcings, `-r` create a realization.
- `--start_date / --end_date` : forcing and realization period (2017-01-01 to 2018-12-31).
- `-o Coralville` : custom output folder name.
- `--source aorc` : use AORC as the meteorological forcing source.

Workflow B: Docker container

This workflow runs the same tool from inside a Docker container as the root user. Because the container starts without the hydrofabric cached, the first run prompts you to download it and to specify the working directory interactively.

Step 1: Enter the data directory

```
cd data/
```

Step 2: Review ngiab-prep options

```
uvx ngiab-prep -h
```

Step 3: Run subset, forcings, and realization

Issue the same core command as in the native workflow:

```
uvx ngiab-prep -i gage-05454300 -sfr \  
  --start_date 2017-01-01 --end_date 2018-12-31 \  
  -o Coralville --source aorc
```

Step 4: Respond to the interactive prompts

On a fresh container the tool walks you through three prompts:

1. **Hydrofabric files are missing — download them now?** Answer y. The tool downloads the hydrofabric (~1.7 GB) plus an additional file (~42.4 MB), then decompresses it.
2. **Output directory is not set — use the default ~/ngiab_preprocess_output/?** Answer n.
3. **Enter the path to the working directory:** /workspace/data/